1 Incompressible Navier-Stokes solver in 3D

We want to solve the following set of dimensionless equations,

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0, \tag{1.1}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{\nabla} \cdot (\boldsymbol{u}\boldsymbol{u}) = -\boldsymbol{\nabla} p + \frac{1}{Re}\nabla^2 \boldsymbol{u} + \boldsymbol{f}_b, \tag{1.2}$$

where $\boldsymbol{f}_b$ indicates additional body forces on the fluid. We will use the pressure-projection method to numerically solve these equations. We first discretize Eq. (1) without the pressure gradient,

$$\frac{\boldsymbol{u}^* - \boldsymbol{u}^n}{\Delta t} = -L_c\Big(\boldsymbol{\nabla} \cdot (\boldsymbol{u}\boldsymbol{u})\Big) + L_v\left(\frac{1}{Re}\nabla^2 \boldsymbol{u}\right) + \boldsymbol{f}_b. \tag{1.3}$$

We indicated the finite difference stencils for the convective term and viscous term as L_c and L_v, respectively. The index n indicates the time level. We now find the following equation for the pressure,

$$\nabla^2 p^{n+1} = \frac{\boldsymbol{\nabla} \cdot \boldsymbol{u}^*}{\Delta t}. \tag{1.4}$$

This gives the final velocity after projection,

$$\frac{\boldsymbol{u}^{n+1} - \boldsymbol{u}^*}{\Delta t} = \boldsymbol{\nabla} p^{n+1}. \tag{1.5}$$

1.1 Staggered grid and spatial stencils

We will perform the simulation on a staggered grid. The domain will be defined as $[0, L_x]$, $[0, L_y]$ and $[0, L_z]$. The pressure will be discretized at the cell centers, such that the positions are defined as follows,

$$\begin{aligned} x &= 0.5\Delta x, 1.5\Delta x, 2.5\Delta x, \ldots, L_x - 0.5\Delta x, \\ y &= 0.5\Delta x, 1.5\Delta x, 2.5\Delta x, \ldots, L_y - 0.5\Delta x, \\ z &= 0.5\Delta x, 1.5\Delta x, 2.5\Delta x, \ldots, L_z - 0.5\Delta x. \end{aligned} \tag{1.6}$$

The velocities are defined at the cell faces along the flow direction,

$$\begin{aligned} x_u &= 0, \Delta x, 2\Delta x, \ldots, L_x, \\ y_v &= 0, \Delta x, 2\Delta x, \ldots, L_y, \\ z_w &= 0, \Delta x, 2\Delta x, \ldots, L_z, \end{aligned} \tag{1.7}$$

where I assumed a square stencil with $\Delta x = \Delta y = \Delta z$. The convective stencil, indicated before as L_c, will be given as follows for the x momentum equation,

$$\begin{aligned} \partial_x \left(u^2\right) &= \frac{\left(u^E\right)^2 - \left(u^W\right)^2}{\Delta x} + \mathcal{O}\left(\Delta x^2\right), \\ \partial_y \left(uv\right) &= \frac{u^N v^N - u^S v^S}{\Delta x} + \mathcal{O}\left(\Delta x^2\right), \\ \partial_z \left(uv\right) &= \frac{u^F w^F - u^B w^B}{\Delta x} + \mathcal{O}\left(\Delta x^2\right). \end{aligned} \tag{1.8}$$

The velocities at the cardinal directions are interpolated as follows,

$$
\begin{aligned}
u^W &= \frac{1}{2}\Big(u_{i-1,j,k} + u_{i,j,k}\Big), \quad u^E = \frac{1}{2}\Big(u_{i,j,k} + u_{i+1,j,k}\Big), \\
u^S &= \frac{1}{2}\Big(u_{i,j-1,k} + u_{i,j,k}\Big), \quad u^N = \frac{1}{2}\Big(u_{i,j,k} + u_{i,j+1,k}\Big), \\
u^B &= \frac{1}{2}\Big(u_{i,j,k-1} + u_{i,j,k}\Big), \quad u^F = \frac{1}{2}\Big(u_{i,j,k} + u_{i,j,k+1}\Big), \\
v^S &= \frac{1}{2}\Big(v_{i-1,j,k} + v_{i,j,k}\Big), \quad v^N = \frac{1}{2}\Big(v_{i-1,j+1,k} + v_{i,j+1,k}\Big), \\
w^B &= \frac{1}{2}\Big(w_{i-1,j,k} + w_{i,j,k}\Big), \quad w^F = \frac{1}{2}\Big(w_{i-1,j,k+1} + w_{i,j,k+1}\Big).
\end{aligned} \tag{1.9}
$$

A similar approach will be used for the y and z momentum equations. The viscous stencil, indicated before as L_v, will be given as follows for the x momentum equation,

$$
\begin{aligned}
\partial_x^2 u &= \frac{u_{i-1,j,k} - 2u_{i,j,k} + u_{i+1,j,k}}{\Delta x^2} + \mathcal{O}\left(\Delta x^2\right), \\
\partial_y^2 u &= \frac{u_{i,j-1,k} - 2u_{i,j,k} + u_{i,j+1,k}}{\Delta x^2} + \mathcal{O}\left(\Delta x^2\right), \\
\partial_z^2 u &= \frac{u_{i,j,k-1} - 2u_{i,j,k} + u_{i,j,k+1}}{\Delta x^2} + \mathcal{O}\left(\Delta x^2\right).
\end{aligned} \tag{1.10}
$$

The gradient of pressure will be computed as follows for the x momentum equation,

$$
\partial_x p = \frac{p_{i,j,k} - p_{i-1,j,k}}{\Delta x} + \mathcal{O}\left(\Delta x^2\right). \tag{1.11}
$$

Finally, the divergence will be computed as follows,

$$
\begin{aligned}
\partial_x u &= \frac{u_{i+1,j,k} - u_{i,j,k}}{\Delta x} + \mathcal{O}\left(\Delta x^2\right), \\
\partial_y v &= \frac{v_{i,j+1,k} - v_{i,j,k}}{\Delta x} + \mathcal{O}\left(\Delta x^2\right), \\
\partial_z w &= \frac{w_{i,j,k+1} - w_{i,j,k}}{\Delta x} + \mathcal{O}\left(\Delta x^2\right).
\end{aligned} \tag{1.12}
$$

1.2 Poisson solver

We will use the Conjugate Gradient Method to solve Eq. (1.4). This is an iterative method that solves the following Matrix equation,

$$
A\boldsymbol{x} = \boldsymbol{b}. \tag{1.13}
$$

Our pressure is denoted by $\boldsymbol{x}$, and the divergence is denoted by $\boldsymbol{b}$. We start by an initial guess for $\boldsymbol{x}$, which is set to the pressure of the previous time step. We then follow with the iterative algorithm 1.1. The stopping criterion $|\boldsymbol{r}_k|_{\max}\Delta t > \varepsilon$ sets a maximum ε of the divergence of the resulting $\boldsymbol{u}^{n+1}$ field. I choose this to be 10^{-6}.

1.3 Boundary conditions

We apply no-sip boundary conditions for the x and y boundaries, and periodic boundary conditions for the z-direction.

1.3.1 Ghost nodes

To do this, we make use of ghost nodes. The scalar fields, in this case pressure, will have ghost nodes at $-0.5\Delta x$ and $L + 0.5\Delta x$ for all directions. The velocity components need to be treated

Algorithm 1.1 The algorithm used to implement the Conjugate Gradient Method

$\boldsymbol{r}_0 \leftarrow \boldsymbol{b} - A\boldsymbol{x}_0$
$\boldsymbol{p}_0 \leftarrow \boldsymbol{r}_0$
$k \leftarrow 0$
while $|\boldsymbol{r}_k|_{\max}\Delta t > \varepsilon$ **do**
  $\alpha_k \leftarrow \frac{|\boldsymbol{r}_k|^2}{\boldsymbol{p}_k \cdot A\boldsymbol{p}_k}$
  $\boldsymbol{x}_{k+1} \leftarrow \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$
  $\boldsymbol{r}_{k+1} \leftarrow \boldsymbol{r}_k - \alpha_k A\boldsymbol{p}_k$
  $\beta_k \leftarrow \frac{|\boldsymbol{r}_{k+1}|^2}{|\boldsymbol{r}_k|^2}$
  $\boldsymbol{p}_{k+1} \leftarrow \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k.$
  $k \leftarrow k + 1$
end while

differently along their flow direction. For instance, we define the ghost nodes for the u velocity to be at $x = -\Delta x$ and $x = L$. Note that $x = L$ is still a physical position inside the fluid. We can still define this node as a ghost node, because the fluid velocity at this node should always be equal to the fluid velocity at $x = 0$. For periodic boundaries, this is trivial. For no-slip boundary conditions, the velocity is zero at both $x = 0$ and $x = L$.

1.3.2 Periodic boundaries

Because we chose $x = L$ to be a ghost node for u, we have the same amount of physical nodes N for both pressure and the velocity components. This means that we can apply the same communication logic to initialize the ghost nodes.

1.3.3 No-slip boundaries

When we impose no-slip on a boundary perpendicular to the flow direction, for instance at $x = 0$, we can simply impose $u(0, y, z) = u_{\text{wall}}$ at these nodes. When we impose no-slip on a boundary along the flow direction, for instance at $y = 0$, we do the following,

$$u\left(x, -\frac{1}{2}\Delta x, z\right) = -u\left(x, \frac{1}{2}\Delta x, z\right) + 2u_{\text{wall}}. \tag{1.14}$$

Finally, we choose $\frac{dp}{dn} = 0$ at no-slip boundaries. We can impose this as follows,

$$p\left(x, -\frac{1}{2}\Delta x, z\right) = p\left(x, \frac{1}{2}\Delta x, z\right). \tag{1.15}$$

Note that we need to compute the Laplacian of the field $\boldsymbol{p}_k$ in the Conjugate Gradient algorithm 1.1. Note that we indicated the pressure as $\boldsymbol{x}$, and $\boldsymbol{p}$ in this case is a correction field. To know what boundary conditions apply to this correction field, we look at the following part of the algorithm,

$$\boldsymbol{x}_{k+1} \leftarrow \boldsymbol{x}_k + \alpha \boldsymbol{p}_k. \tag{1.16}$$

From this, we get the following,

$$\frac{\partial}{\partial n}\boldsymbol{x}_{k+1} \leftarrow \frac{\partial}{\partial n}\boldsymbol{x}_k + \alpha \frac{\partial}{\partial n}\boldsymbol{p}_k. \tag{1.17}$$

So if both the normal gradient of the initial pressure $\boldsymbol{x}_k$ and next pressure $\boldsymbol{x}_{k+1}$ are zero, then the gradient of $\boldsymbol{p}$ must also be zero. This means that we can use the same boundary condition for both the pressure field and the correction field used in the Conjugate Gradient Method when we apply no-slip boundaries.

1.4 Implementation

A 3D, MPI-parallelized implementation of the algorithm I described can be found on GitHub [1]. For guidance during parameter selection, we can define a CFL number,

$$\mathrm{CFL} = \frac{\Delta t}{\Delta x} \max\Big(u + v + w\Big). \tag{1.18}$$

Generally, the CFL number is kept below 0.5.

2 Results

2.1 Testing Poisson solver

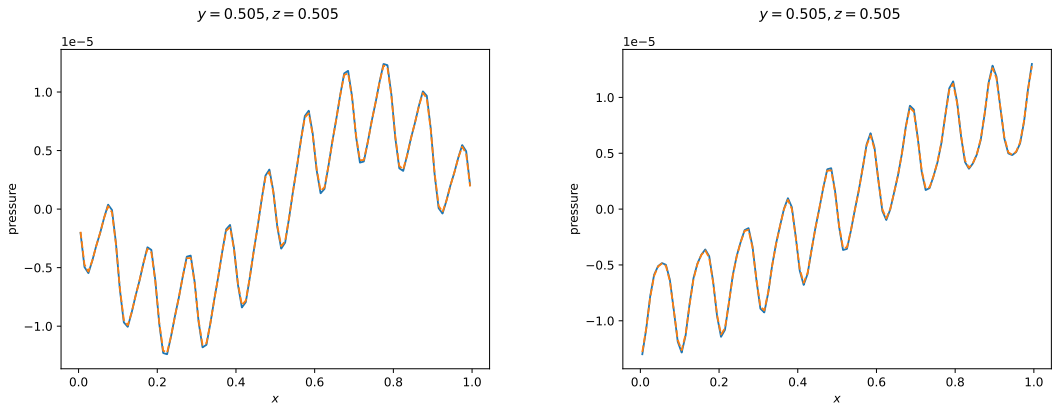
I will test the Poisson solver with periodic boundary conditions by setting the right hand side, indicated by $b(x, y, z)$, of the Poisson equation equal to the following,

$$b(x,y,z) = \sin\left(\frac{2\pi x}{L_x}\right)\sin\left(\frac{2\pi y}{L_y}\right)\sin\left(\frac{2\pi z}{L_z}\right) + \frac{1}{2}\sin\left(\frac{20\pi x}{L_x}\right)\sin\left(\frac{20\pi y}{L_y}\right)\sin\left(\frac{20\pi z}{L_z}\right) + \frac{1}{10}\sin\left(\frac{40\pi x}{L_x}\right)\sin\left(\frac{40\pi y}{L_y}\right)\sin\left(\frac{40\pi z}{L_z}\right). \tag{2.1}$$

To test the no-slip boundary conditions we have the following,

$$b(x,y,z) = \cos\left(\frac{\pi x}{L_x}\right)\cos\left(\frac{\pi y}{L_y}\right)\cos\left(\frac{\pi z}{L_z}\right) + \frac{1}{2}\cos\left(\frac{19\pi x}{L_x}\right)\cos\left(\frac{19\pi y}{L_y}\right)\cos\left(\frac{19\pi z}{L_z}\right) + \frac{1}{10}\cos\left(\frac{39\pi x}{L_x}\right)\cos\left(\frac{39\pi y}{L_y}\right)\cos\left(\frac{39\pi z}{L_z}\right). \tag{2.2}$$

Note that higher frequencies are added to be more challenging to the Poisson solver. If we only included the largest wavelengths, the solver would converge in 1 step. It now takes about 3 steps when we use $\Delta t = 1$ in our stopping criterion described in Sec. (1.2). The results can be found in Fig. 2.1.



(a) Periodic

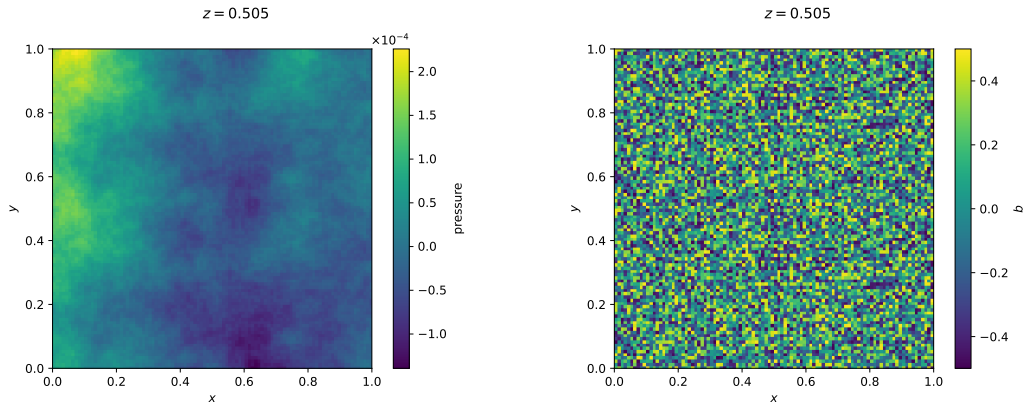
(b) No-slip

Figure 2.1: Numerical (blue) versus analytical (orange) solutions to Poisson equation along line in 3D domain.

We can stress-test the solver, forcing it to take more iterations to converge, by setting b equal to random values in the range $[-0.5, 0.5]$. This is shown in Fig. (2.2). This takes 461 steps to converge to our strict convergence criterion.

2.2 Poiseuille flow

A simple test case, before we move on to the lid-driven cavity flow, can be the Poiseuille flow. We have periodic boundary conditions in the z-direction, with a length of 1. We have periodic boundary conditions in the x-direction, also with a length of 1. We have no-slip boundary conditions in the y-direction, with a height of 64. When introducing a constant force of 1 in the x-direction, we obtain a parabolic flow. The results can be shown in Fig. (2.3)



(a) Pressure solution

(b) Right-hand side

Figure 2.2: The solution for pressure (left) and the right hand side of the Poisson equation (right).

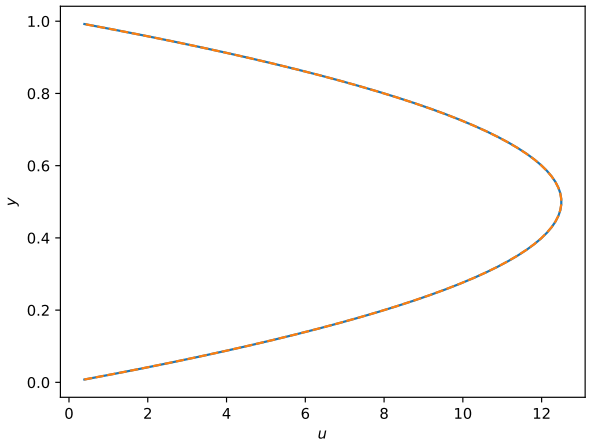
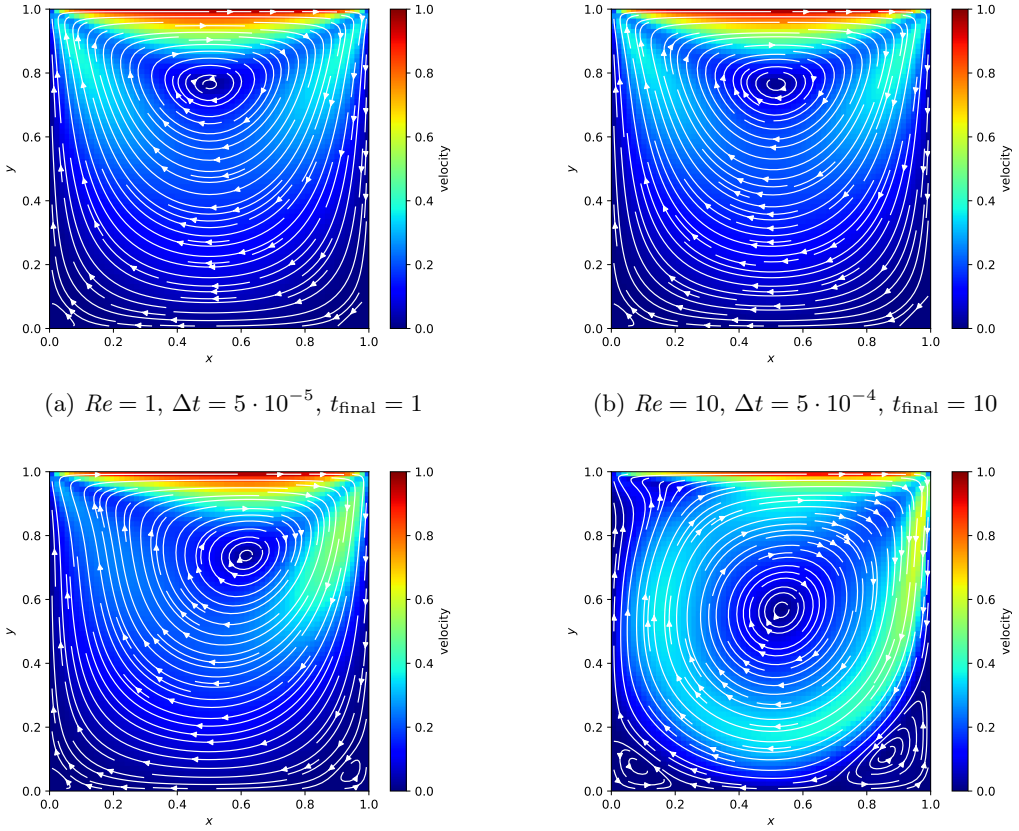


Figure 2.3: The numerical flow profile (blue) and the analytical profile (orange)

2.3 Lid-driven Cavity Flow

For the lid-driven cavity flow, we have periodic boundary conditions in the z-direction, and no-slip boundary conditions in the x and y directions. The flow is driven by moving the top wall with a horizontal velocity of 1. We can run simulations on a 64 by 64 grid for $Re = 1, 10, 100$ and 1000. We show the velocity profiles with correspoding streamlines for these Reynolds numbers in Fig. (2.4). The timesteps Δt are chosen to be as large as they could without introducing instability. The final times t_{final} are chosen at the point where the simulation reaches equilibrium. To keep the CFL number below 0.5, we don't increase Δt further than $5 \cdot 10^{-3}$ to prevent CFL numbers higher than 0.5.



(a) $Re = 1$, $\Delta t = 5 \cdot 10^{-5}$, $t_{\text{final}} = 1$

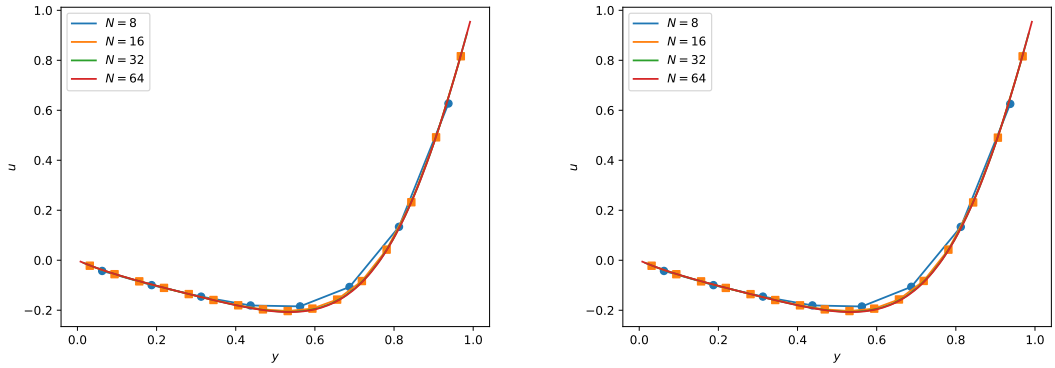
(b) $Re = 10$, $\Delta t = 5 \cdot 10^{-4}$, $t_{\text{final}} = 10$

(c) $Re = 100$, $\Delta t = 5 \cdot 10^{-3}$, $t_{\text{final}} = 100$

(d) $Re = 1000$, $\Delta t = 5 \cdot 10^{-3}$, $t_{\text{final}} = 1000$

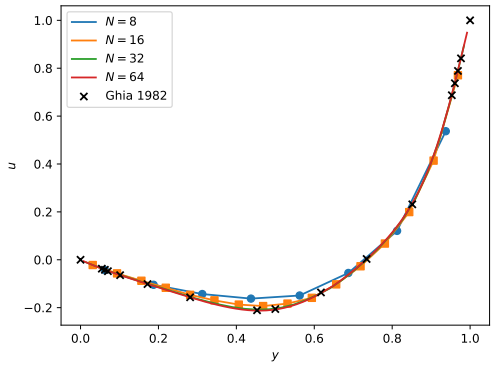
Figure 2.4: Velocity plots with streamlines given for different Reynolds numbers.

We can plot the u velocity at $x = \frac{1}{2}L$ and v at $y = \frac{1}{2}L$. The results are shown in Figures 2.5 and 2.6. We have comparison data from previous research that shows good agreement for our numerical solutions at $Re = 100$ and $Re = 1000$ [2].

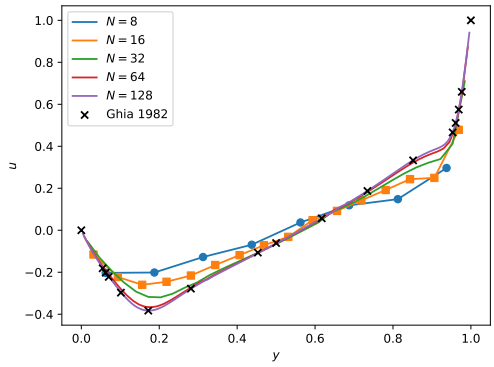


(a) $Re = 1$

(b) $Re = 10$

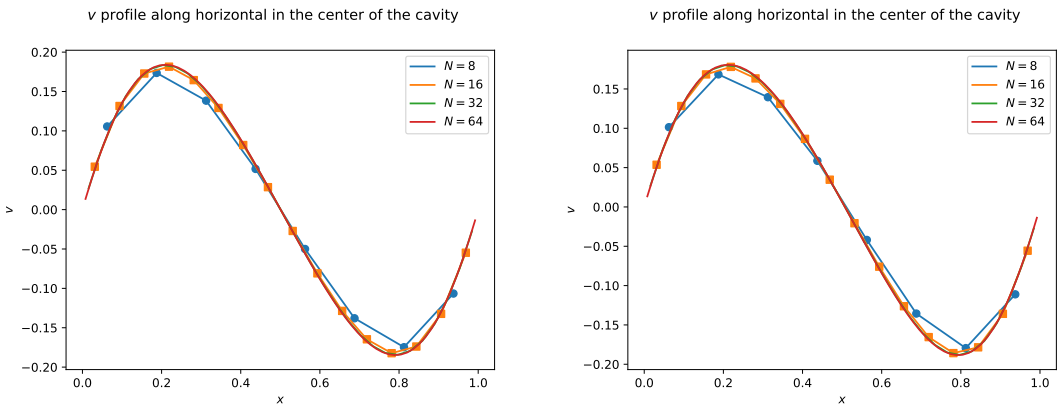


(c) $Re = 100$



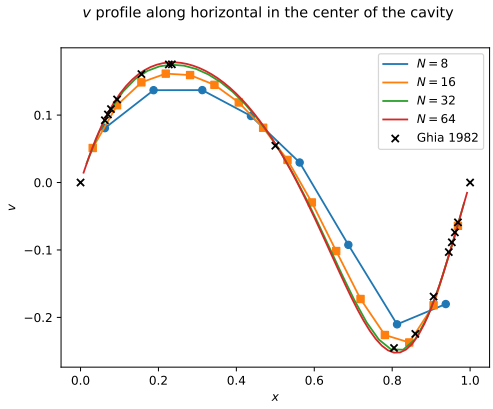
(d) $Re = 1000$

Figure 2.5: Line plots of u velocity through vertical center of domain for different Reynolds numbers.

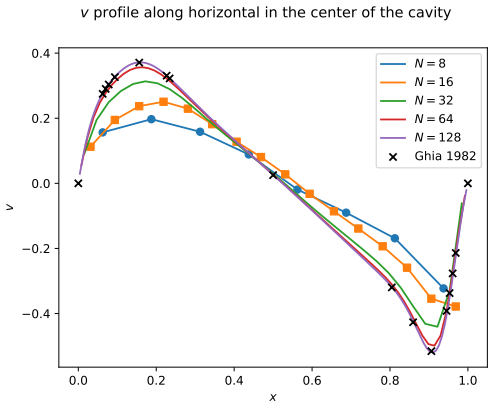


(a) $Re = 1$

(b) $Re = 10$



(c) $Re = 100$



(d) $Re = 1000$

Figure 2.6: Line plots of v velocity through horizontal center of domain for different Reynolds numbers.

References

[1] D. Schrijver, *CFD*, https://github.com/dschrijver/CFD, GitHub repository, 2026.

[2] U. Ghia, K. N. Ghia, and C. T. Shin, Journal of Computational Physics **48**, 387 (1982).